DATA ANALYTICS

Complete Learning Guide

From vague business requests to trustworthy, decision-ready insights

Designed for self-paced learning. This guide uses explanations, worked examples, workplace scenarios, common pitfalls and knowledge checks—not just definitions.

Learning path: question → data → preparation → analysis → visualisation → recommendation

Platform assessment guidance: 70% required to pass; retakes are available. Focus on understanding why an answer is correct.

How to use this guide

Study one module at a time. For each concept ask: What does it mean? When would I use it? What could make the result misleading? Attempt the checkpoint questions before reading the suggested answers.

Learning habit	How to apply it
Start with a decision	Ask what someone will do differently after seeing the analysis.
Write a measurable question	Define population, metric, comparison and time period.
Check data grain	Confirm what one row represents before counting or joining.
Document choices	Record exclusions, assumptions, cleaning rules and caveats.
Explain plainly	State the finding, evidence, limitation and next action.

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01	The Analytics Mindset
02	Data Sources & Structure
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Module 1: The Analytics Mindset

Turning vague requests into answerable questions.

What you should be able to do

• Convert a broad request into a measurable question.

• Distinguish descriptive, diagnostic, predictive and prescriptive analysis.

• Identify the decision an analysis supports.

Start with the decision

A request like “sales are bad—analyse the data” is too vague. Ask who needs the answer, what decision they must make, which outcome matters, and when the decision is needed. Design the analysis backwards from the decision.

Make the question measurable

A good question names the population, metric, time window and comparison. For example: “How did weekly revenue in each region change over the last eight weeks compared with the previous eight weeks, and which categories contributed most?”

Four types of analysis

Descriptive asks what happened. Diagnostic asks why it happened. Predictive estimates what may happen next. Prescriptive recommends an action given goals and constraints. They can be used in sequence, but they are not interchangeable.

Metrics and denominators

A metric is a quantified measure tracked over time. Conversion rate could be purchases divided by eligible website sessions. Define the numerator, denominator, population and time window. A percentage without its denominator can mislead.

Worked example

A store receives 1,000 visits and 50 purchases. Conversion = 50 ÷ 1,000 × 100 = 5%. Summarising the rate is descriptive; comparing it by device to investigate a drop is diagnostic; forecasting next month is predictive; recommending a checkout experiment is prescriptive.

Workplace scenario

Scenario: A manager says “support is too slow.” Clarify whether this means first response time, time to resolution, or both. Specify ticket types and dates, compare with a target, then decide whether the analysis will support staffing or process changes.

Common mistakes to avoid

• Confusing correlation with a proven cause.

• Reporting percentages without denominators.

• Choosing an easy metric rather than one tied to the decision.

Check your understanding

Q1. What happened to revenue last month?

Q2. What is missing from “Are customers happy?”

Q3. A model estimates next month’s cancellations. Which analysis type?

Suggested answers

A1. Descriptive.

A2. A defined customer group, satisfaction measure, time period and possibly a comparison.

A3. Predictive.

Remember: Good analysis clarifies a decision, uses fit-for-purpose data and communicates uncertainty honestly.

Module 2: Data Sources & Structure

Where data lives and what shape it arrives in.

What you should be able to do

• Describe structured, semi-structured and unstructured data.

• Explain rows, columns, keys and granularity.

• Judge whether a source can answer the question.

Structured data

Structured data follows a defined schema, usually rows and columns: orders, invoices, employee records or sales tables. It is easy to filter, join and aggregate with SQL or spreadsheets.

Semi-structured data

JSON responses, XML and event logs contain labels or markers but may have nested or optional fields. Inspect their structure before flattening them into tables.

Unstructured data

Reviews, emails, images, audio and video do not naturally fit a fixed table. They may require text processing, transcription, image recognition or manual coding.

Rows, columns and keys

A row is a record at a chosen level of detail; a column is an attribute. A primary key uniquely identifies a row. A customer ID can repeat in an orders table because one customer can place many orders.

Granularity

Granularity is what one row represents: one order, order item, customer or day. If an order total is repeated on three item rows, summing it across those rows triples revenue.

Source suitability

Check coverage, freshness, fields, definitions, quality, permissions and time period. Record source lineage and transformations.

Workplace scenario

Example: Customers(customer_id, region) has one row per customer. Orders(order_id, customer_id, amount) has one row per order. Joining on customer_id allows revenue analysis by region. Before counting, ask what one row means and which key identifies it.

Common mistakes to avoid

• Assuming every ID is unique in every table.

• Assuming absent fields mean an event did not happen.

• Joining tables without checking one-to-many relationships.

Check your understanding

Q1. Is a JSON API response usually structured, semi-structured or unstructured?

Q2. What does one row per order item mean?

Q3. Why can a join multiply rows?

Suggested answers

A1. Semi-structured.

A2. Each row represents an item line, not necessarily an entire order.

A3. One row on one side can match several rows on the other side.

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Module 3: Data Cleaning & Preparation

The work that determines whether the answer is right.

What you should be able to do

• Handle missing values, duplicates and inconsistent formats.

• Choose and document imputation approaches.

• Investigate outliers before removing them.

Profile first

Inspect row counts, types, missingness, duplicate keys, unique values, date ranges and impossible values before editing. Preserve a raw copy.

Missing values

Missing may mean unknown, not collected, not applicable or lost. These meanings differ. Dropping every incomplete row can bias results if missingness is concentrated in a group.

Imputation

Imputation replaces a missing value with an estimate. Mean imputation can be distorted by extremes; median is often more robust for skewed data. A category such as “Unknown” may be safer for missing labels. For predictive modelling, fit imputation rules on training data only.

Duplicates and formats

Exact duplicate rows differ from legitimate repeated events. Define a business key and investigate before removing records. Standardise dates, currencies, units, spelling and whitespace.

Outliers

An outlier may be a typo, fraud, a bulk order or a genuine rare case. Validate against source systems and business rules. Correct confirmed errors, keep valid extremes, and document exclusions.

Reproducibility

Record the issue, rule, number of affected rows and reason. Example: “12 negative quantities were checked against source records; 9 corrected and 3 excluded with approval.”

Workplace scenario

Worked example: observed ages are 21, 22, missing, 24 and 78. Mean of observed values is 36.25; median is 23. The extreme value pulls the mean upward. Median imputation may be more representative, but first determine why age is missing and whether imputation is justified.

Common mistakes to avoid

• Deleting outliers without investigating.

• Replacing all missing values with zero even when zero has meaning.

• Making undocumented manual edits.

• Using full-data statistics before a train/test split.

Check your understanding

Q1. When can median imputation be preferable to mean?

Q2. Is a repeated customer ID necessarily a duplicate?

Q3. What should happen before removing a very large transaction?

Suggested answers

A1. When values are skewed or have extremes, if imputation is appropriate.

A2. No; customers can have multiple transactions.

A3. Validate it against the source and business context.

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Module 4: Spreadsheet Modelling

Lookups, pivots and workbooks that do not break.

What you should be able to do

• Build reliable lookup and conditional logic.

• Summarise data with pivot tables.

• Use references and validation to make workbooks auditable.

Structure the workbook

Separate raw data, calculations and outputs. Label inputs, document assumptions, use consistent formats and avoid hidden hard-coded values.

Lookup functions

XLOOKUP or VLOOKUP retrieves a value from another table using a shared key. Prefer stable identifiers over names that can be duplicated or misspelled. Handle unmatched keys explicitly.

Conditional logic

IF tests a condition and returns a result for true or false. AND and OR combine conditions. Test boundary values such as exactly 70 and just below 70.

Absolute references

A relative reference such as A2 changes when copied. An absolute reference such as B2 locks row and column. Use absolute references for fixed targets or rates.

Pivot tables

Pivot tables aggregate by groups. Put Region in Rows, Month in Columns, and Revenue in Values using Sum. Choose Count for record counts and Average for averages. Reconcile totals against the source.

Validation and controls

Data validation restricts entries to allowed values or ranges. Include units, error checks and reconciliation totals so another person can trace outputs.

Workplace scenario

Formula: `=IF(B2>=$F$1,"Target met","Below target")`. When copied down, B2 changes to B3, B4, while F1 remains fixed. To compare monthly revenue by region, use Region as Rows, Month as Columns, and Sum of Revenue as Values.

Common mistakes to avoid

• Using names instead of stable IDs for lookups.

• Using COUNT when SUM is needed.

• Forgetting to lock a fixed target cell.

• Assuming a pivot table refreshes automatically.

Check your understanding

Q1. What does B2 do when copied?

Q2. Which aggregation gives total revenue?

Q3. Why handle unmatched lookup values?

Suggested answers

A1. Keeps column B and row 2 fixed.

A2. SUM.

A3. Otherwise missing matches can be mistaken for zero or misclassified.

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Module 5: SQL Thinking

Asking a database precise questions.

What you should be able to do

• Write selection, filtering and aggregation queries.

• Understand WHERE, GROUP BY and HAVING.

• Choose joins and validate row counts.

SELECT and FROM

SELECT chooses columns or expressions; FROM identifies the source table. Explicit columns are clearer than SELECT * in reports.

WHERE

WHERE filters rows before grouping. Use IS NULL rather than = NULL. Date filters should respect the data type and time zone.

Aggregation and GROUP BY

COUNT, SUM, AVG, MIN and MAX summarise values. GROUP BY produces a result for each grouping combination.

HAVING

HAVING filters groups after aggregation. WHERE might select 2026 orders; GROUP BY computes revenue per customer; HAVING keeps customers above a revenue threshold.

Joins

INNER JOIN returns matches on both sides. LEFT JOIN keeps all left-side rows and adds right-side matches when present. One-to-many joins expand rows, which can be correct for detail analysis but dangerous for totals.

Validate queries

Check expected output grain, row counts, unmatched keys and control totals. Test on a small sample and use meaningful aliases.

Workplace scenario

Example query:

```
SELECT customer_id, SUM(amount) AS revenue
FROM Orders
WHERE order_date >= '2026-01-01'
AND order_date < '2027-01-01'
GROUP BY customer_id;
```

The exclusive end date includes all timestamps on 31 December.

Common mistakes to avoid

• Using WHERE to filter an aggregate result instead of HAVING.

• Joining on a non-unique name.

• Ignoring NULL and unmatched keys.

• Assuming joined row counts equal source row counts.

Check your understanding

Q1. Which clause filters rows before grouping?

Q2. Which join preserves every left-side row?

Q3. What does COUNT(*) count after a join?

Suggested answers

A1. WHERE.

A2. LEFT JOIN.

A3. Rows in the joined result, not necessarily unique original entities.

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Module 6: Descriptive Statistics

Summarising data without being misled by averages.

What you should be able to do

• Choose mean, median and mode appropriately.

• Interpret range and standard deviation.

• Recognise skew and the limits of correlation.

Centre

Mean is sum divided by count; median is the middle sorted value; mode is the most frequent value/category. Mean uses all values but is sensitive to extremes; median is more robust for skew.

Spread

Range is maximum minus minimum. Variance is based on squared distances from the mean; standard deviation is its square root and uses the original unit. Spread describes variability, not accuracy.

Skew

A right-skewed distribution has a long right tail, often caused by a few large values. Income, order value and waiting time may be right-skewed; mean can be above median.

Correlation

Correlation describes the strength and direction of a relationship. Positive variables tend to move together; negative variables move in opposite directions. Correlation does not prove causation.

Compare fairly

Interpret averages with sample size, spread, population and period. If data is skewed, report the median or distribution as well as the mean.

Workplace scenario

Example: Values 10, 11, 12, 13, 54 have mean 20 and median 12. The 54 pulls the mean upward. Two teams may have the same average resolution time but different standard deviations, meaning one team is less consistent.

Common mistakes to avoid

• Using a mean without checking skew or outliers.

• Claiming correlation proves causation.

• Comparing different populations or time windows.

• Interpreting standard deviation without units or context.

Check your understanding

Q1. Which centre measure is more robust to an extreme value?

Q2. Same mean but different standard deviation means what?

Q3. Does strong correlation prove causation?

Suggested answers

A1. Median.

A2. Their spread or consistency differs.

A3. No. Confounders and other explanations may exist.

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Module 7: Exploratory Data Analysis

Finding patterns while checking what could mislead you.

What you should be able to do

• Run a structured first pass on unfamiliar data.

• Use segmentation to explain aggregate changes.

• Understand cohorts and Simpson’s paradox.

EDA workflow

Start with the question and data dictionary. Check shape, types, missingness, duplicates and ranges. Summarise variables, plot distributions, examine relationships and segment important outcomes. EDA generates hypotheses; it does not automatically prove them.

Segmentation

Split the population by region, channel, product, device or tenure to reveal differences hidden in an aggregate. Choose segments related to the question.

Cohorts

A cohort shares a starting event or period, such as customers who signed up in January. Track retention or repeat purchase over time. Compare equal follow-up windows.

Simpson’s paradox

A trend visible within each subgroup can reverse when groups are combined, often because the groups have different sizes or difficulty levels. Check group mix before explaining an aggregate trend.

Signal and noise

Small changes may be random, especially in small samples. Consider effect size, sample size, consistency over time and a plausible mechanism. Avoid repeatedly slicing data until a desired pattern appears.

Workplace scenario

Example: Overall conversion falls from 5% to 4%. Segmenting by device may reveal mobile conversion is stable while desktop conversion falls. Traffic mix can also change the overall rate, so report both within-device rates and the mix.

Common mistakes to avoid

• Exploring without a question.

• Comparing cohorts with unequal follow-up.

• Treating every segment difference as meaningful.

• Reporting only an aggregate when group mix changed.

Check your understanding

Q1. Why segment conversion by device?

Q2. What defines a cohort?

Q3. How can Simpson’s paradox mislead?

Suggested answers

A1. To find group differences and whether mix explains the overall change.

A2. A shared starting event or period, tracked over time.

A3. Combined trends can reverse subgroup trends when group weights differ.

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Module 8: Dashboard & Visualisation Design

Charts that answer the question at a glance.

What you should be able to do

• Choose chart types for analytical tasks.

• Organise a dashboard around decision priority.

• Reduce clutter and design honest, accessible charts.

Match chart to task

Bars compare categories; lines show change over ordered time; scatterplots examine relationships between numeric variables; histograms show distributions. Pie charts work best for a few parts of a meaningful whole.

Bar and line charts

Bar charts should usually start at zero because bar length encodes magnitude. Line charts suit time series; avoid connecting unrelated categories as if they were continuous.

KPI tiles

Show the headline number plus period, target or comparison. “Revenue ■12 lakh” is less useful than “■12 lakh this month, 8% above last month and 3% below target.”

Data-ink and clutter

Decorative 3D effects, unnecessary gridlines and repeated labels can distract. Remove decoration that adds no meaning, but prioritise clarity over minimalism for its own sake.

Honest and accessible charts

Label axes, units, dates and definitions. Truncated bar axes can exaggerate differences. Do not rely on colour alone; consider contrast, labels and text size.

Dashboard hierarchy

Put decision-relevant KPIs first, followed by trends and diagnostic detail. Show data freshness, metric definitions and clear filters.

Workplace scenario

Chart choice: compare regional revenue → bar chart; weekly revenue trend → line chart; discount versus order value → scatterplot; customer age distribution → histogram. A useful dashboard might show revenue, conversion, orders, target variance, weekly trend, and regional breakdown.

Common mistakes to avoid

• Using pie charts with many slices.

• Omitting units and periods from KPIs.

• Using colour as the only distinction.

• Truncating axes to exaggerate differences.

Check your understanding

Q1. Best chart for monthly revenue trend?

Q2. What context should a KPI tile include?

Q3. Why can a truncated bar axis mislead?

Suggested answers

A1. Line chart.

A2. Metric definition, period, and useful comparison or target.

A3. It can make small differences look much larger.

Remember: Good analysis clarifies a decision, uses fit-for-purpose data and communicates uncertainty honestly.

Module 9: Storytelling With Data

Getting a decision made, not just a chart admired.

What you should be able to do

• Structure findings for a non-technical audience.

• State assumptions and limitations honestly.

• Write a concise, actionable executive summary.

Finding versus insight

A finding says what the data shows. An insight explains why it matters. “Mobile conversion is 2.8%” is a finding; “mobile conversion is far below desktop and most visits are mobile, so checkout usability may be a high-value opportunity” is closer to an insight, if the comparison is valid.

Executive summary

Answer the question first. Include the main conclusion, key evidence, recommendation and important limitation. The summary should stand alone without requiring every chart.

Assumptions and limitations

An assumption is a condition treated as true, such as each order ID representing one completed order. A limitation restricts what the evidence can establish, such as missing cash transactions.

Confidence statements

Explain whether evidence is strong, suggestive or incomplete and what could change the conclusion. Use “consistent with” when causation has not been established.

Actionable recommendations

Specify an action, owner, time horizon and success measure when possible. Separate observed evidence from explanations and actions that still need testing.

Workplace scenario

Example: Weak: “Sales dropped by 9%. See charts.” Stronger: “Revenue fell 9% versus the previous four weeks, mainly because completed orders declined in two regions; average order value was stable. Investigate checkout errors this week. The analysis shows where the decline occurred but does not yet prove its cause.”

Common mistakes to avoid

• Starting with methods rather than the answer.

• Using unexplained jargon.

• Hiding uncertainty in footnotes.

• Recommending actions unsupported by evidence.

Check your understanding

Q1. How does an insight differ from a finding?

Q2. When should you avoid saying a change caused an outcome?

Q3. Name three parts of a useful executive summary.

Suggested answers

A1. An insight explains why a finding matters for a decision.

A2. When the analysis only shows association or lacks causal evidence.

A3. Conclusion, supporting evidence, recommendation, and limitation/next step.

Remember: Good analysis clarifies a decision, uses fit-for-purpose data and communicates uncertainty honestly.

Final Assessment-Style Practice Quiz

Choose one best answer for each question. This practice quiz is not the platform's official assessment. A score of 14/20 equals 70%.

1. A manager asks how many orders arrived last month.

A. Predictive

B. Descriptive

C. Prescriptive

D. Diagnostic

2. A team compares churn across plan types to investigate a rise.

A. Descriptive

B. Diagnostic

C. Prescriptive

D. Sampling

3. Which is the strongest measurable question?

A. Why is the app bad?

B. Are users happy?

C. How did 30-day retention for Q2 signups compare with Q1?

D. Inspect everything

4. Which is usually semi-structured?

A. Relational table

B. JPEG image

C. JSON API response with optional fields

D. Printed poster

5. One row per order item describes the table's:

A. Colour

B. Granularity

C. Correlation

D. Variance

6. A primary key should:

A. Repeat for each customer

B. Uniquely identify each row

C. Always be a name

D. Contain no numbers

7. For strongly right-skewed data, which centre measure is more robust?

A. Median

B. Maximum

C. Range

D. Standard deviation

8. Before removing a very large transaction:

A. Delete it

B. Replace with mean

C. Check source and context

D. Hide it

9. Which reference stays fixed when copied?

A. A1

B. A1

C. A$1 locks both dimensions

D. A1 always stays fixed

10. Best spreadsheet tool for revenue by region?

A. Spell check

B. Pivot table

C. Merge cells

D. Text wrap

11. Which SQL clause filters rows before grouping?

A. HAVING

B. ORDER BY

C. WHERE

D. SELECT DISTINCT

12. Which join preserves all left-table rows?

A. INNER

B. LEFT

C. CROSS only

D. SELF

13. Revenue doubles after a one-to-many join. Likely issue?

A. Font size

B. Row multiplication at different grain

C. Median

D. Line chart

14. Correlation between ad spend and sales proves ads caused sales to rise.

A. True

B. False

15. Why use cohort analysis?

A. Remove missing values

B. Track a group sharing a starting event over time

C. Equalise averages

D. Guarantee causality

16. Best chart for a monthly trend?

A. Line

B. Pie with 30 slices

C. 3D chart

D. Word cloud

17. A KPI tile should include:

A. Only a big number

B. Metric, period and useful context

C. No units

D. Decoration

18. An insight differs from a finding because it:

A. Never uses data

B. Explains why a finding matters

C. Must be a pie chart

D. Always proves causation

19. Feature users have higher retention but self-selected. Best conclusion?

A. Feature definitely caused it

B. Feature is useless

C. Association exists; test causality with stronger design

D. Delete users

20. Best first step for “sales are bad—analyse data”?

A. Build dashboard immediately

B. Clarify decision and measurable question

C. Delete outliers

D. Choose colours

Answer Key and Explanations

Q	Answer	Why
1	B	It summarises what happened.
2	B	It investigates why the outcome changed.
3	C	It defines outcome, population, period and comparison.
4	C	JSON has labels and structure, often with nested/optional fields.
5	B	Granularity is what one row represents.
6	B	It identifies a row uniquely.
7	A	The median is less affected by extremes.
8	C	An outlier may be genuine or an error.
9	B	Dollar signs lock row and column.
10	B	Pivot tables group and aggregate.
11	C	WHERE filters individual rows.
12	B	LEFT JOIN keeps every left-side row.
13	B	Repeated order-level amounts may be summed several times.
14	B	Confounders and other explanations may exist.
15	B	Cohorts enable time-aligned behaviour comparisons.
16	A	Line charts show change over ordered time.
17	B	Context makes the number interpretable.
18	B	It connects evidence to implications or action.
19	C	Self-selection may explain the difference.
20	B	A clear question guides the entire analysis.

Quick Revision Sheet

Concept	Remember
Descriptive	What happened? Counts, totals, rates and trends.
Diagnostic	Why might it have happened? Compare segments and test explanations.
Predictive	What is likely next? Estimate future outcomes with uncertainty.
Prescriptive	What action should be considered, given goals and constraints?
Metric	A defined measure with numerator, denominator, population and time window.
Granularity	What one row represents; establish it before joins or counts.
Cleaning	Profile first; investigate missingness, duplicates, formats and outliers.
Spreadsheets	Use stable keys, correct aggregation, locked references, validation and reconciliation.
SQL	WHERE filters rows; GROUP BY forms groups; HAVING filters groups; joins may multiply rows.
Statistics	Median resists extremes; standard deviation describes spread; correlation is not causation.
EDA	Inspect quality, distributions, relationships and segments; patterns are hypotheses.
Visualisation	Choose the chart for the task; label units, period, baseline and context.
Storytelling	Lead with the answer, show evidence, recommend action and state limitations.

Before submitting an analysis

- Can I state the decision and measurable question clearly?
- Do I know the grain, keys, time coverage and limitations of each source?
- Have I checked missing values, duplicates, formats and outliers?
- Can I reconcile totals and explain the query/formula logic?
- Have I checked important segments and changes in population mix?
- Does every chart have a clear title, units, time period and baseline?
- Does my summary separate evidence, assumptions, interpretation and recommendation?

Final principle: A polished dashboard cannot repair a poorly defined question or unreliable data. Clear thinking, careful validation and honest communication are the foundations of good analytics.